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Spectral Causality: Causal Direction Estimation via Magnetic Laplacians and Hodge Decomposition

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Abstract

We introduce *spectral causality*, a framework for causal direction estimation via the *magnetic Laplacian*, an Hermitian matrix whose complex-valued eigenvectors encode edge directionality as phase. We define a *Directional Predictability Index* (DPI) that resolves the circularity of earlier utility-based formulations: under additive noise model assumptions, DPI consistently estimates causal direction without domain knowledge. Connecting the framework to *Hodge decomposition*, we prove that gradient–curl–harmonic separation of edge flows yields a causal potential recovering topological order under directed acyclic graph (DAG) structure, and establish *scale invariance* of the gradient energy ratio r_{gradient}, showing that causal structure emergence depends on knowledge *quality* rather than quantity. Experiments on synthetic random DAGs (n=5–20, N=200–1000), the Sachs protein signaling network (n=11, 17 ground-truth edges), and the UCI Heart Disease dataset (N=297, 5 variables) demonstrate competitive recovery against DirectLiNGAM, the PC algorithm, GES, and NOTEARS. On Sachs data with partial knowledge, spectral causality achieves a true positive rate (TPR) of 0.56 and a false discovery rate (FDR) of 0.27, while providing unique DAG-adequacy diagnostics and detecting feedback loops invisible to DAG-based methods.

Keywords: Causal discovery, Magnetic Laplacian, Hodge decomposition, Spectral graph theory, Directed graphs

1 Introduction

Causal inference—determining whether X causes Y—is a central problem in science and medicine. The dominant approaches include structural equation models (SEMs) with the *do*-calculus (Pearl, 2009), potential outcomes (Rubin, 1974), the linear non-Gaussian acyclic model (LiNGAM) (Shimizu et al., 2006), and Granger causality for time series (Granger, 1969).

This paper proposes a fundamentally different principle: *reading causal directions from the spectral structure (eigenvalues and eigenvectors) of a graph Laplacian.* We call this approach **spectral causality**.

Basic idea. Given n variables $\{X_1, \ldots, X_n\}$ with putative causal relations represented as a weighted directed graph $G = (V, E, w)$, the *magnetic Laplacian* $\mathcal{L}^{(q)}$ is an Hermitian matrix whose eigenvectors are complex-valued. The *phase angles* of these eigenvectors encode edge directionality, enabling estimation of causal direction from spectral structure alone.

Key distinction from DAG-based methods. LiNGAM and related methods assume a directed acyclic graph (DAG). In biomedical systems, feedback loops are ubiquitous (e.g., inflammation $\to$ organ damage $\to$ inflammation). Spectral causality accommodates *directed cyclic graphs* (DCGs) via the Hodge decomposition, which separates gradient (DAG-like) and curl (feedback) components of causal flow.

Contributions. The main contributions of this paper are:

1. **Directional Predictability Index (DPI).** A data-driven asymmetric statistic that enables causal direction estimation at $\alpha = 0$ (no domain knowledge), resolving the circularity in earlier utility-based formulations (Section 4).

2. **DAG-free causal inference.** Via Hodge decomposition, we separate DAG-compatible flow (gradient) from feedback (curl), providing the gradient energy ratio r_{gradient} as a quantitative measure of DAG adequacy (Section 5).

3. **Scale invariance theorem.** We prove that r_{gradient} depends only on the *structure* (sign pattern) of the asymmetric component of the utility matrix, not on its magnitude—establishing that knowledge *quality*, not quantity, governs causal structure emergence (Section 8).

4. **Ensemble Causal Direction (ECD).** A pipeline integrating LiNGAM's local identifiability with spectral causality's global structural analysis, achieving broad coverage of Hill's nine epidemiological criteria (Section 9).

5. **Partial identifiability of DPI.** Under the additive noise model (ANM), the ANM component of DPI consistently identifies causal direction as $n \to \infty$, providing a non-circular theoretical foundation (Section 10).

6. **Multi-dataset validation.** Experiments on synthetic random DAGs ($n = 5$–20), the Sachs protein signaling network (Sachs et al., 2005; $n = 11$, 17 ground-truth edges), and the UCI Heart Disease dataset ($N = 297$, 5 variables) demonstrate competitive structural recovery against DirectLiNGAM, the PC algorithm, GES, and NOTEARS (Section 7).

Figure 1 provides a conceptual overview of the three complementary approaches unified in this work.

Paper outline. Section 2 reviews graph Laplacians. Section 3 introduces the magnetic Laplacian. Section 4 formulates spectral causality and DPI. Section 5 develops the Hodge decomposition connection. Section 6 surveys related work. Section 7 presents experiments. Section 8 analyzes the DAG phase transition. Section 9 describes the ECD pipeline. Section 10 addresses identifiability. Section 11 discusses limitations and future work.

2 Preliminaries: Graph Laplacians

Definition 1 (Graph Laplacian) *Let $G = (V, E, w)$ be a weighted undirected graph with $|V| = n$ and weight function $w : E \to \mathbb{R}_{>0}$. The* weighted adjacency matrix $W \in \mathbb{R}^{n\times n}$*, the*

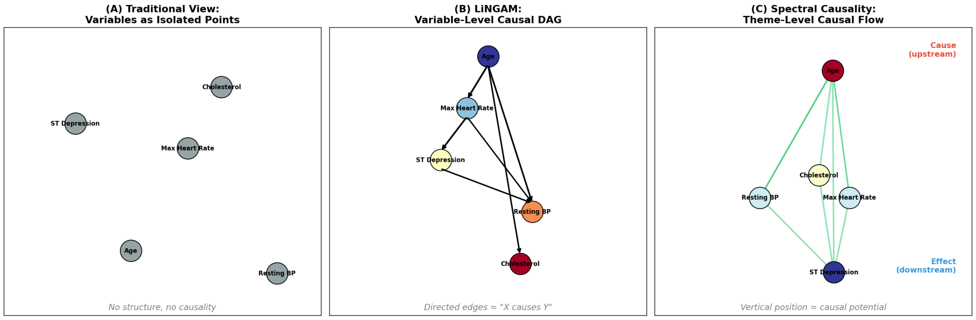

Figure 1: Three complementary approaches to causal inference: (1) structural equation models and DAG-based methods (e.g., LiNGAM), (2) spectral graph methods via the magnetic Laplacian, and (3) Hodge-theoretic flow decomposition. Spectral causality unifies (2) and (3).

degree matrix $D = \mathrm{diag}(d_1, \ldots, d_n)$ *with* $d_i = \sum_j W_{ij}$, *and the* graph Laplacians *are:*

$$L = D - W \quad (\textit{unnormalized}), \qquad \mathcal{L} = I - D^{-1/2} W D^{-1/2} \quad (\textit{normalized}).$$

Proposition 2 (Basic properties) *The following hold:*

(i) L *is symmetric positive semidefinite with eigenvalues* $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_n$.

(ii) *The eigenvector for* $\lambda_1 = 0$ *is* $\mathbf{1} = (1, \ldots, 1)^\top$.

(iii) $\lambda_2 > 0$ *if and only if* G *is connected (algebraic connectivity / Fiedler value).*

(iv) *For any* $f \in \mathbb{R}^n$, $f^\top L f = \sum_{(i,j)\in E} w_{ij}(f_i - f_j)^2 \geq 0$.

Proof Property (iv) follows by expanding the quadratic form of $L = D - W$. Property (i) is immediate from (iv). Property (ii) holds since $L\mathbf{1} = \mathbf{0}$. Property (iii) is the algebraic connectivity theorem. ■

Property (iv) is fundamental: $f^\top L f$ is small when f assigns similar values to adjacent nodes. Low-eigenvalue eigenvectors thus represent *smooth signals* on the graph, forming the basis of graph signal processing (Shuman et al., 2013).

Remark 3 (Limitation of undirected Laplacians) *Since* $L = D - W$ *is* symmetric, *it cannot distinguish the direction* $i \to j$ *from* $j \to i$. *The directed Laplacian* $L_d = D_{\mathrm{out}} - W$ *is generally non-symmetric with complex eigenvalues, making spectral analysis difficult. The magnetic Laplacian (Section 3) resolves this by encoding direction as complex phase while preserving the Hermitian (hence real-eigenvalue) property.*

3 The Magnetic Laplacian

3.1 Physical Motivation

The magnetic Laplacian originates in quantum mechanics. For a charged particle in a magnetic field $\mathbf{B}$ with vector potential $\mathbf{A}$, the Hamiltonian is $H = (\mathbf{p} - e\mathbf{A})^2/2m$. A particle traversing a closed loop acquires the Aharonov–Bohm phase $\exp\big(i \oint \mathbf{A} \cdot d\mathbf{r}\big)$, whose *direction dependence* can be repurposed to encode graph edge directions.

3.2 Definition

Definition 4 (Magnetic Laplacian) *Let $G = (V, E, w)$ be a weighted directed graph and $q \in [0, 0.5]$ a* charge parameter*. Define the* Hermitian adjacency matrix $H^{(q)} \in \mathbb{C}^{n \times n}$ *by*

$$H_{ij}^{(q)} = w_{ij} \cdot \exp\big(i \cdot 2\pi q \cdot \sigma_{ij}\big), \tag{1}$$

where $w_{ij} = (w_{ij}^{\text{orig}} + w_{ji}^{\text{orig}})/2$ is the symmetrized weight, and $\sigma_{ij} \in \{-1, 0, +1\}$ is the direction sign:

$$\sigma_{ij} = \begin{cases} +1 & \text{if } i \to j, \\ -1 & \text{if } j \to i, \\ 0 & \text{if no edge.} \end{cases}$$

The normalized magnetic Laplacian *is*

$$\mathcal{L}^{(q)} = I - D^{-1/2} H^{(q)} D^{-1/2}, \tag{2}$$

where $D = \operatorname{diag}(d_1, \dots, d_n)$ with $d_i = \sum_j |H_{ij}^{(q)}|$.

Proposition 5 (Properties of the magnetic Laplacian) *The following hold:*

(i) *$H^{(q)}$ is Hermitian: $H_{ji}^{(q)} = \overline{H_{ij}^{(q)}}$.*

(ii) *$\mathcal{L}^{(q)}$ is Hermitian positive semidefinite with real, non-negative eigenvalues.*

(iii) *The eigenvectors of $\mathcal{L}^{(q)}$ are generally* complex-valued*.*

(iv) *At $q = 0$, $\mathcal{L}^{(0)}$ reduces to the standard normalized Laplacian $\mathcal{L}$ (no directional information).*

Proof For (i): since $w_{ji} = w_{ij}$ and $\sigma_{ji} = -\sigma_{ij}$,

$$H_{ji}^{(q)} = w_{ij} \cdot \exp(-i \cdot 2\pi q \cdot \sigma_{ij}) = \overline{w_{ij} \cdot \exp(i \cdot 2\pi q \cdot \sigma_{ij})} = \overline{H_{ij}^{(q)}}.$$

Property (ii) follows from (i) and the construction $\mathcal{L}^{(q)} = I - D^{-1/2} H^{(q)} D^{-1/2}$. The Hermitian property ensures real eigenvalues; positive semidefiniteness follows from the quadratic form $f^* \mathcal{L}^{(q)} f \geq 0$ for all $f \in \mathbb{C}^n$. Property (iii) is a consequence of the complex-valued entries of $H^{(q)}$. Property (iv) holds since $\exp(0) = 1$ yields real entries. ∎

3.3 The Charge Parameter q

The parameter q controls directional sensitivity:

q	Phase $2\pi q$	Effect
0	0	No direction. $\exp(0) = 1$; real matrix.
0.25	$\pi/2$	Maximum sensitivity. $e^{i\pi/2} = i$, $e^{-i\pi/2} = -i$.
0.5	π	Direction reversal. $e^{i\pi} = -1$.

Remark 6 *At $q = 0.25$, for an edge $i \to j$: $H_{ij}^{(q)} = i \cdot w_{ij}$ and $H_{ji}^{(q)} = -i \cdot w_{ij}$, achieving maximal separation of forward and reverse directions via the imaginary unit.*

3.4 Phase Angles and Directionality

Each eigenvector $u_k \in \mathbb{C}^n$ of $\mathcal{L}^{(q)}$ admits the polar decomposition

$$u_k(j) = |u_k(j)| \cdot \exp\bigl(i \cdot \theta_k(j)\bigr),$$

where $|u_k(j)|$ is the *amplitude* and $\theta_k(j) = \arg(u_k(j))$ the *phase angle* at node j.

Theorem 7 (Phase–direction correspondence) *For $q > 0$ and a utility-directed graph with consistent causal ordering (Definition 8), the phase angles $\theta_k(j)$ of low-frequency eigenvectors separate causal upstream nodes (sources) from downstream nodes (sinks) in the complex plane. Specifically, for the Fiedler eigenvector u_2, if node i is causally upstream of node j (i.e., $\phi(i) > \phi(j)$ in the Hodge causal potential of Definition 20), then $\theta_2(i)$ and $\theta_2(j)$ tend to occupy distinct angular sectors.*

Proof The key mechanism is phase accumulation along directed paths. For a directed edge $i \to j$ at $q = 0.25$, the Hermitian adjacency contributes $H_{ij}^{(q)} = iw_{ij}$ and $H_{ji}^{(q)} = -iw_{ij}$. In the eigenequation $\mathcal{L}^{(q)} u_k = \lambda_k u_k$, the asymmetric phase factors force a strictly positive phase increment $\Delta\theta > 0$ per directed edge.

We prove this in two stages in Appendix A. *Stage 1* (Lemma 36): For a path graph $v_1 \to \cdots \to v_n$, the normalized magnetic Laplacian eigenequation yields the recursion $(1 - \lambda)u(v_k) = \frac{-iw}{d_{v_k}} u(v_{k-1}) + \frac{iw}{d_{v_k}} u(v_{k+1})$. Writing $u(v_k) = r_k e^{i\theta_k}$ and taking the real part shows that $1 - \lambda_2 > 0$ and $r_k > 0$ require $\theta_{k+1} - \theta_k > 0$, giving monotone phase ordering. *Stage 2* (Theorem 37): For tree DAGs, we use structural induction on depth. The base case (star graph) gives $\theta_2(c_j) = \theta_2(r) + \pi/2$ for each child c_j of root r. The inductive step combines this with the subtree hypothesis via transitivity. A perturbation argument extends the result to non-uniform weights $w_{ij} > 0$. ∎

4 Spectral Causality: Formulation

4.1 The Utility Directed Graph

Definition 8 (Utility directed graph) *Given n variables* $\{X_1, \ldots, X_n\}$*, define a* utility function $U : \{1, \ldots, n\}^2 \to \mathbb{R}_{\geq 0}$ *where* $U(i,j)$ *quantifies "how useful information about* X_i *is for reasoning about* X_j*." The* utility directed graph $G_U = (V, E, w, \sigma)$ *has:*

- *Symmetrized weight* $w(i,j) = \big(U(i,j) + U(j,i)\big)/2$*,*
- *Direction sign* $\sigma(i,j) = \operatorname{sign}\big(U(i,j) - U(j,i)\big)$*.*

4.2 The Directional Predictability Index (DPI)

A critical limitation of using $|\hat{\rho}_{ij}|$ (absolute correlation) as the data-driven component is that $|\hat{\rho}_{ij}| = |\hat{\rho}_{ji}|$, rendering the utility matrix symmetric at $\alpha = 0$. This makes direction estimation impossible without external knowledge—a *circularity* in the original formulation.

Definition 9 (Directional Predictability Index) *For observed data* $\mathbf{X} \in \mathbb{R}^{N \times n}$*, the* DPI matrix $D_{\mathrm{DPI}} \in \mathbb{R}^{n \times n}$ *is defined by*

$$D_{\mathrm{DPI}}(i \to j) = |\hat{\rho}_{ij}| \cdot \big(1 + \gamma \cdot \bar{A}(i,j)\big), \tag{3}$$

where $\gamma > 0$ *is a directional strength parameter (we use* $\gamma = 1$*) and* $\bar{A}(i,j)$ *is the mean of three normalized asymmetric statistics:*

$$\bar{A}(i,j) = \tfrac{1}{3}\big[\hat{A}_{\mathrm{reg}}(i,j) + \hat{A}_{\mathrm{ANM}}(i,j) + \hat{A}_{\mathrm{ent}}(i,j)\big]. \tag{4}$$

The three components are:

(i) ***Regression coefficient asymmetry*** $\hat{A}_{\mathrm{reg}}$*: the normalized difference* $|\beta_{j|i}| - |\beta_{i|j}|$ *of unstandardized regression coefficients, exploiting the fact that* $|\beta_{j|i}| \neq |\beta_{i|j}|$ *when* $\mathrm{Var}(X_i) \neq \mathrm{Var}(X_j)$*.*

(ii) ***Additive noise model (ANM) residual independence*** $\hat{A}_{\mathrm{ANM}}$*: for each pair* (i,j)*, fit* $X_j = \beta X_i + \varepsilon$ *and evaluate independence of* $\hat{\varepsilon}$ *and* X_i *via the Hilbert–Schmidt Independence Criterion (HSIC) with median-heuristic kernel bandwidth. Lower HSIC implies* $X_i \to X_j$ *is more plausible.*

(iii) ***Conditional entropy reduction*** $\hat{A}_{\mathrm{ent}}$*: the asymmetry in* $H(X_j) - H(X_j|X_i)$ *versus* $H(X_i) - H(X_i|X_j)$*, estimated via k-nearest-neighbor entropy estimators.*

Each component is normalized to $[-1, 1]$*.*

Proposition 10 (Asymmetry of DPI) $D_{\mathrm{DPI}}(i \to j) \neq D_{\mathrm{DPI}}(j \to i)$ *whenever* $\bar{A}(i,j) \neq 0$*.*

Proof Since each component satisfies $\hat{A}_k(j,i) = -\hat{A}_k(i,j)$ (antisymmetry of normalized directional statistics), we have $\bar{A}(j,i) = -\bar{A}(i,j)$. Therefore,

$$D_{\mathrm{DPI}}(j \to i) = |\hat{\rho}_{ij}|(1 - \gamma\bar{A}(i,j)) \neq |\hat{\rho}_{ij}|(1 + \gamma\bar{A}(i,j)) = D_{\mathrm{DPI}}(i \to j)$$

whenever $\bar{A}(i,j) \neq 0$. ∎

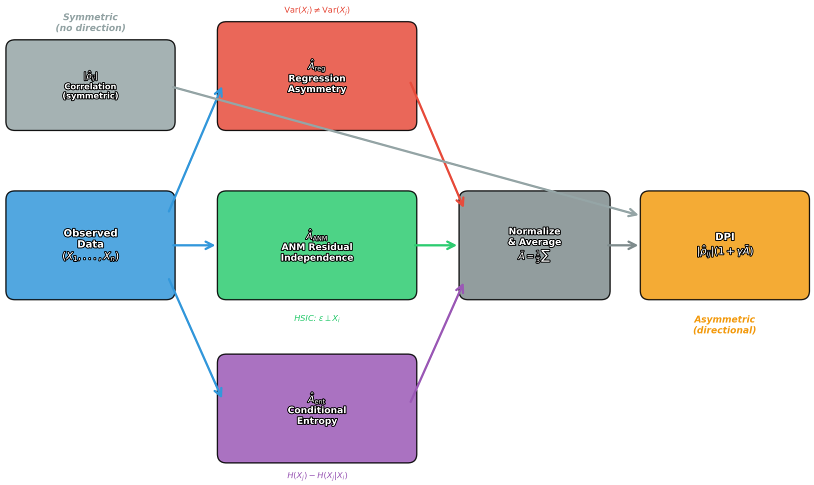

Figure 2: Architecture of the Directional Predictability Index (DPI). Three independent asymmetric statistics—regression coefficient asymmetry $\hat{A}_{\mathrm{reg}}$, ANM residual independence $\hat{A}_{\mathrm{ANM}}$, and conditional entropy reduction $\hat{A}_{\mathrm{ent}}$—are normalized and averaged to produce $\bar{A}(i,j)$, which modulates the symmetric correlation $|\hat{\rho}_{ij}|$ into the asymmetric DPI score. The modular design allows each component to be replaced or extended independently.

Hybrid utility function. The full utility function combines domain knowledge and data:

$$U(i,j) = \alpha \cdot C_{\mathrm{domain}}(i,j) + (1-\alpha) \cdot D_{\mathrm{DPI}}(i \to j), \tag{5}$$

where $\alpha \in [0,1]$ is the domain knowledge mixing ratio and $C_{\mathrm{domain}}(i,j) \in [0,1]$ is a *domain knowledge matrix* encoding the degree to which variable X_i is believed to causally influence variable X_j, based on external knowledge sources such as expert elicitation, published literature, or structured ontologies. Unlike a binary adjacency matrix, C_{domain} permits graded confidence: $C_{\mathrm{domain}}(i,j) = 0.8$ encodes strong believed influence, while $C_{\mathrm{domain}}(i,j) = 0.1$ encodes weak or uncertain influence. The asymmetry $C_{\mathrm{domain}}(i,j) \neq C_{\mathrm{domain}}(j,i)$ encodes directional prior knowledge. Crucially, at $\alpha = 0$ the DPI asymmetry preserves directional signal, unlike the $|\hat{\rho}|$-based formulation. Figure 2 illustrates this modular architecture.

4.3 Spectral Causal Coupling and Direction

Definition 11 (Spectral Causal Coupling) *Given the eigendecomposition $\mathcal{L}^{(q)} = U\Lambda U^*$ with $U = [u_1, \ldots, u_n]$, $\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_n)$, the* Spectral Causal Coupling *between nodes i and j is*

$$\mathrm{SCC}(i,j) = \sum_{k=1}^{n} f(\lambda_k) \cdot |u_k(i)| \cdot |u_k(j)| \cdot \cos\big(\theta_k(i) - \theta_k(j)\big), \tag{6}$$

where $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ *is an eigenvalue weight function (typically* $f(\lambda) = \lambda$*) and* $\theta_k(i) = \arg(u_k(i))$.

Definition 12 (Spectral Causal Direction) *The* Spectral Causal Direction *is*

$$\mathrm{SCD}(i, j) = \sum_{k=1}^{n} f(\lambda_k) \cdot |u_k(i)| \cdot |u_k(j)| \cdot \sin\bigl(\theta_k(i) - \theta_k(j)\bigr). \tag{7}$$

Proposition 13 (Symmetry properties) (i) $\mathrm{SCC}(i, j) = \mathrm{SCC}(j, i)$ (symmetric).

(ii) $\mathrm{SCD}(i, j) = -\mathrm{SCD}(j, i)$ (skew-symmetric).

(iii) $\mathrm{SCD}(i, i) = 0$.

Proof (i) follows from $\cos(\alpha - \beta) = \cos(\beta - \alpha)$. (ii) follows from $\sin(\alpha - \beta) = -\sin(\beta - \alpha)$. (iii) is immediate from (ii). ■

$\mathrm{SCD}(i, j) > 0$ suggests causal direction $i \to j$; $\mathrm{SCD}(i, j) < 0$ suggests the reverse.

Theorem 14 (Complex Causal Index) *Define the* Complex Causal Index

$$\mathrm{CCI}(i, j) = \sum_{k=1}^{n} f(\lambda_k) \cdot |u_k(i)| \cdot |u_k(j)| \cdot \exp\bigl(i(\theta_k(i) - \theta_k(j))\bigr). \tag{8}$$

Then $\mathrm{SCC}(i, j) = \mathrm{Re}[\mathrm{CCI}(i, j)]$ *and* $\mathrm{SCD}(i, j) = \mathrm{Im}[\mathrm{CCI}(i, j)]$.

Proof Direct application of Euler's formula $\exp(i\alpha) = \cos\alpha + i \sin\alpha$. ■

Remark 15 (Geometric interpretation) *In the complex plane,* $|\mathrm{CCI}(i, j)|$ *measures causal coupling strength and* $\arg(\mathrm{CCI}(i, j))$ *measures causal direction (Figure 3).*

4.4 Properties of the SCD Matrix

Proposition 16 (SCD matrix properties) *The matrix* $S \in \mathbb{R}^{n \times n}$ *with* $S_{ij} = \mathrm{SCD}(i, j)$ *satisfies:*

(i) $S = -S^\top$ (skew-symmetric).

(ii) $\mathrm{tr}(S) = 0$.

(iii) *If* $q = 0$*, then* $S = O$ *(the zero matrix).*

Proof (i) and (ii) follow from Proposition 13. For (iii), at $q = 0$ the eigenvectors are real, so $\theta_k(j) \in \{0, \pi\}$ for all k, j, whence $\sin(\theta_k(i) - \theta_k(j)) = 0$. ■

Property (iii) is important: spectral causality *requires* directional information ($q > 0$). This parallels LiNGAM's requirement of non-Gaussianity.

Definition 17 (Spectral causal score) *The* spectral causal score *of node* i *is* $s(i) = \sum_{j \neq i} \mathrm{SCD}(i, j)$. *Larger* $s(i)$ *indicates a more upstream (causal) position. By skew-symmetry,* $\sum_i s(i) = 0$.

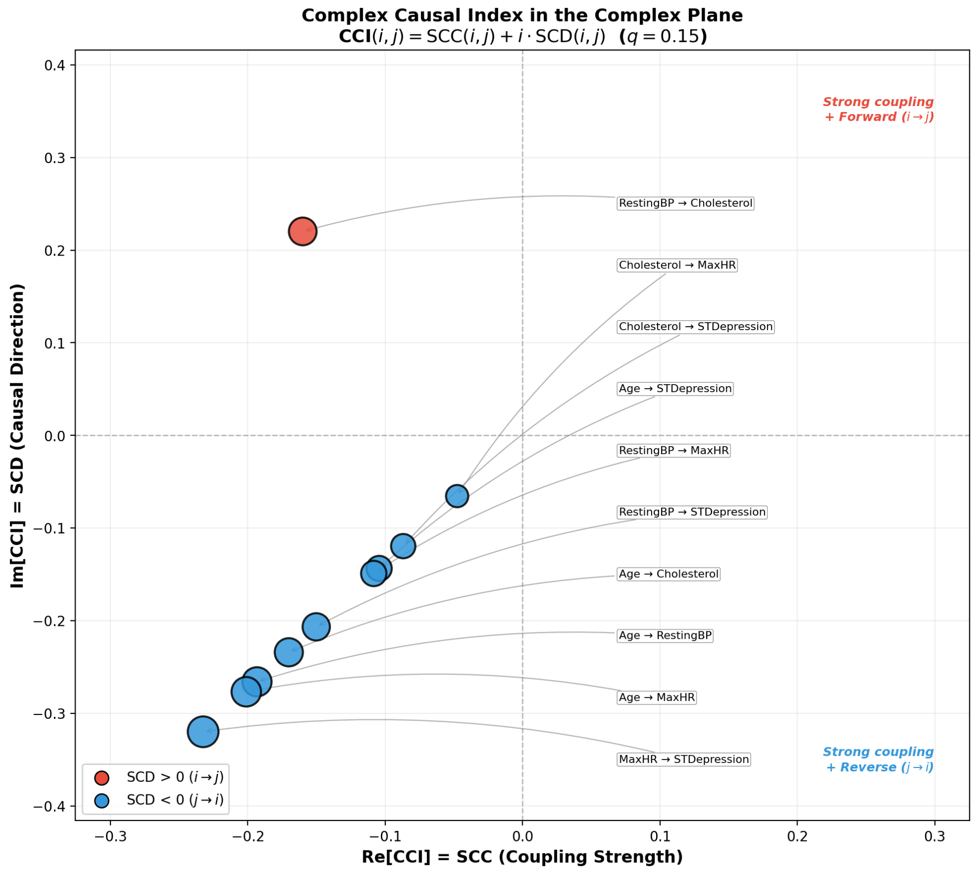

Figure 3: Complex Causal Index $\mathrm{CCI}(i, j)$ for all variable pairs plotted in the complex plane ($q = 0.15$). The real axis (SCC) represents coupling strength; the imaginary axis (SCD) represents causal direction. Red points indicate forward direction ($i \to j$); blue points indicate reverse direction. All SCC values are negative, indicating inhibitory coupling dominates; the SCD spread reveals a clear directional hierarchy with Age as the upstream root node.

5 Hodge Decomposition and Causal Flow

5.1 Differential Forms on Graphs

Definition 18 (Cochain complex) *For a directed graph $G = (V, E)$, define:*

- $C^0 = \mathbb{R}^{|V|}$ *(node functions),* $\quad C^1 = \mathbb{R}^{|E|}$ *(edge flows).*
- *Coboundary* $\delta_0 : C^0 \to C^1$*:* $(\delta_0 f)(i \to j) = f(j) - f(i)$ *(discrete gradient).*
- *Coboundary* $\delta_1 : C^1 \to C^2$*: curl on triangles.*

5.2 The Hodge Decomposition Theorem

Theorem 19 (Hodge decomposition on graphs (Jiang et al., 2011; Lim, 2020)) *Any edge flow $\omega \in C^1$ decomposes uniquely as*

$$\omega = \underbrace{\delta_0 \phi}_{gradient} + \underbrace{\delta_1^* \psi}_{curl} + \underbrace{h}_{harmonic}, \tag{9}$$

where δ_1^* is the adjoint of δ_1, and the three components are mutually orthogonal: $\langle \delta_0 \phi, \delta_1^* \psi \rangle = 0$, etc.

The causal interpretation of each component is:

Component	Mathematical meaning	Causal interpretation
$\delta_0\phi$ (gradient)	Potential-driven flow	**Causal flow** (DAG-like, unidirectional)
$\delta_1^*\psi$ (curl)	Local circulation	**Feedback loops** (local interactions)
h (harmonic)	Global circulation	**Homeostatic regulation** (systemic)

5.3 Causal Potential

Definition 20 (Causal potential) *The* causal potential $\phi : V \to \mathbb{R}$ *is the potential function in the gradient component $\delta_0 \phi$, obtained by solving*

$$L\phi = \delta_0^* \omega, \tag{10}$$

which is a Poisson equation on the graph. Since L is positive semidefinite, ϕ is unique up to an additive constant.

Proposition 21 (Causal potential recovers topological sort) *If ω represents a pure DAG flow (i.e., the curl and harmonic components are zero), then the ordering of nodes by ϕ coincides with a topological sort of the DAG.*

Proof When $\omega = \delta_0 \phi$ (pure gradient), for every directed edge $i \to j$ we have $\omega(i \to j) = \phi(j) - \phi(i) > 0$, hence $\phi(j) > \phi(i)$. This is precisely the topological ordering condition. ■

Definition 22 (Gradient energy ratio) *The* gradient energy ratio *is*

$$r_{\text{gradient}} = \frac{\|\delta_0 \phi\|^2}{\|\omega\|^2}, \tag{11}$$

which measures the proportion of total flow energy captured by the DAG-compatible (gradient) component. $r_{\text{gradient}} \approx 1$ indicates DAG adequacy; $r_{\text{gradient}} \ll 1$ indicates dominant feedback structure.

6 Related Work

Magnetic Laplacian on directed graphs. Fanuel and Suykens (2017) pioneered applying deformed Laplacians to spectral ranking in directed networks, connecting to PageRank via group synchronization. Fanuel et al. (2017) used magnetic eigenmaps for community detection. de Resende and da Costa (2020) characterized large directed networks via the magnetic Laplacian spectrum, showing that q progressively captures cyclic structure. Zhang et al. (2022) proposed MagNet, a GNN based on the magnetic Laplacian. These works establish the magnetic Laplacian's ability to encode directionality; our contribution is its application to *causal inference*.

Hodge decomposition in networks. Jiang et al. (2011) applied Hodge decomposition to statistical ranking, using the gradient component for global ranking and the curl for intransitivity. Maehara and Ohkawa (2025) applied Hodge decomposition (ddHodge) to single-cell RNA-seq data, reconstructing gene expression dynamics on data manifolds. Their demonstration that developmental processes are governed by potential landscapes directly supports the biological plausibility of our causal potential.

Signal processing on DAGs. Seifert et al. (2023) defined causal Fourier analysis on DAGs under a "few causes" sparsity assumption. Misiakos et al. (2024) extended this to time-series graph data. Stanković et al. (2024) proposed graph zero-padding to enable Fourier transforms on DAGs (whose adjacency matrices have all-zero eigenvalues). Our approach is complementary: these methods recover signals given a known DAG; we *estimate* causal direction from spectral structure.

Continuous DAG learning. NOTEARS (Zheng et al., 2018) reformulated the acyclicity constraint as a continuous function $\mathrm{tr}(e^{W \circ W}) - d = 0$, enabling gradient-based structure learning. GOLEM (Ng et al., 2020) improved optimization efficiency. M'Charrak et al. (2025) introduced a Fiedler-eigenvalue-based connectivity constraint for DAG learning. Our spectral causality encodes direction via the magnetic Laplacian's *phase*, while M'Charrak et al. use the standard Laplacian's *smallest nonzero eigenvalue* for connectivity.

Information-theoretic causality. Transfer entropy (Schreiber, 2000) quantifies directed information flow in time series. Convergent cross mapping (Sugihara et al., 2012) detects causality in weakly coupled dynamical systems. Both require time-series data; spectral causality applies to cross-sectional snapshots, but the utility asymmetry ("knowing X_i informs about X_j") is conceptually analogous to transfer entropy.

LiNGAM and medical applications. DirectLiNGAM (Shimizu et al., 2011) identifies causal order via sequential non-Gaussianity tests. Kotoku et al. (2020) applied it to Osaka health checkup data ($n \approx 10^5$), and Okuda et al. (2023) proposed workflow-constrained longitudinal LiNGAM. Our ECD pipeline (Section 9) uses LiNGAM output as initialization for spectral causality.

LLMs and causal inference. Le et al. (2024) proposed multi-agent causal discovery (MAC). Sheth et al. (2024) evaluated LLMs on causal queries. These works motivate LLM-based construction of the domain knowledge matrix C_{domain} as a future direction.

Table 1: Domain knowledge matrix C_{domain} for the UCI Heart Disease variables. Entry (i, j) represents the believed causal influence strength of variable i on variable j, based on cardiovascular physiology. Asymmetry $C(i, j) \neq C(j, i)$ encodes directional prior knowledge.

$C(i, j)$	Age	RestBP	Chol	MaxHR	STDep
Age	—	0.6	0.4	0.5	0.3
RestBP	0.1	—	0.2	0.3	0.4
Cholesterol	0.1	0.3	—	0.1	0.3
MaxHR	0.1	0.2	0.1	—	0.5
STDep	0.0	0.1	0.0	0.2	—

7 Experiments

7.1 Data and Variables

We use the UCI Heart Disease Dataset (Cleveland subset; Detrano et al., 1989), selecting five continuous clinical variables:

$$\mathbf{x} = (X_1, X_2, X_3, X_4, X_5) = (\text{Age, RestBP, Chol, MaxHR, STDep}).$$

Sample size $N = 297$; all variables are standardized to zero mean and unit variance.

Domain knowledge matrix. We construct C_{domain} from established cardiovascular physiology (Table 1). Each entry $C_{\text{domain}}(i, j) \in [0, 1]$ represents the strength of the believed causal influence of X_i on X_j, elicited from medical literature (Detrano et al., 1989). For example, $C_{\text{domain}}(\text{Age}, \text{RestBP}) = 0.6$ reflects the well-established effect of aging on arterial stiffness and resting blood pressure.

7.2 LiNGAM Baseline

Applying DirectLiNGAM (Shimizu et al., 2011), the estimated causal order is $X_1 \prec X_4 \prec X_5 \prec X_2 \prec X_3$ (Age $\to$ MaxHR $\to$ STDep $\to$ RestBP $\to$ Chol), with principal causal effects: $B_{42} = -0.395$ (Age $\to$ MaxHR, age-related decline in maximum heart rate), $B_{21} = +0.309$ (Age $\to$ RestBP, age-related blood pressure increase), $B_{54} = -0.348$ (MaxHR $\to$ STDep, exercise-intolerance-induced ischemia).

7.3 Magnetic Laplacian Eigenvectors

We construct the utility directed graph (Eq. 5) with $\alpha = 0.6$, combining C_{domain} (Table 1) at 60% weight with data-driven DPI at 40% weight, and compute $\mathcal{L}^{(q)}$ for $q \in \{0,\ 0.1,\ 0.25\}$.

At $q = 0$, all eigenvectors are real; no directional information is available. At $q = 0.25$, the eigenvectors become complex-valued with informative phase angles (Table 2).

Figure 4 shows the complex-plane distribution of eigenvector components, demonstrating how increasing q separates upstream from downstream nodes.

Table 2: Phase angles of the Fiedler eigenvector (u_2) at $q = 0.25$. Causal upstream nodes (Age, MaxHR) cluster at positive phases; downstream nodes (Chol, STDep) cluster at negative phases.

Variable	$\lvert u_2 \rvert$ (amplitude)	θ_2 (phase, degrees)
Age	0.53	0.0°
Resting BP	0.35	164.6°
Cholesterol	0.42	−84.3°
Max HR	0.47	34.7°
ST Depression	0.44	−40.6°

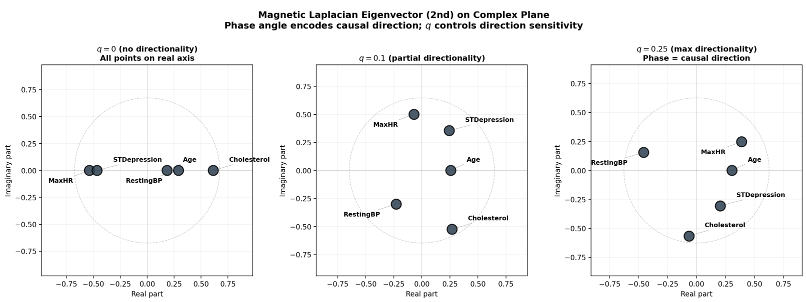

Figure 4: Fiedler eigenvector of the magnetic Laplacian plotted in the complex plane for $q = 0$, 0.1, and 0.25. At $q = 0$, all points lie on the real axis. As q increases, variables spread into the complex plane, with phase angle ordering reflecting causal flow direction.

7.4 Hodge Decomposition Results

Applying Hodge decomposition to the edge flow $\omega(i, j) = w(i, j) \cdot \sigma(i, j)$:

$$\begin{aligned} \|\delta_0 \phi\|^2 / \|\omega\|^2 &= 85.9\% \quad \text{(gradient = DAG-like causal flow)}, \\ \|\delta_1^* \psi\|^2 / \|\omega\|^2 &= 14.1\% \quad \text{(curl = feedback loops)}. \end{aligned}$$

$r_{\text{gradient}} = 85.9\%$ indicates predominantly DAG-like structure (Figure 5).

The causal potential ϕ orders the variables as Age > Chol > RestBP $\approx$ MaxHR > STDep (Table 3), placing Age as the most upstream and ST Depression as the most downstream, which is clinically sensible.

7.5 Synthetic Data Benchmark

To evaluate spectral causality against established methods under controlled conditions, we generate synthetic datasets from known causal structures and compare structural recovery performance.

Table 3: Causal potential ϕ from Hodge decomposition ($\alpha = 0.6$).

Rank	Variable	ϕ
1	Age	0.000
2	Cholesterol	-0.168
3	Resting BP	-0.204
4	Max Heart Rate	-0.204
5	ST Depression	-0.324

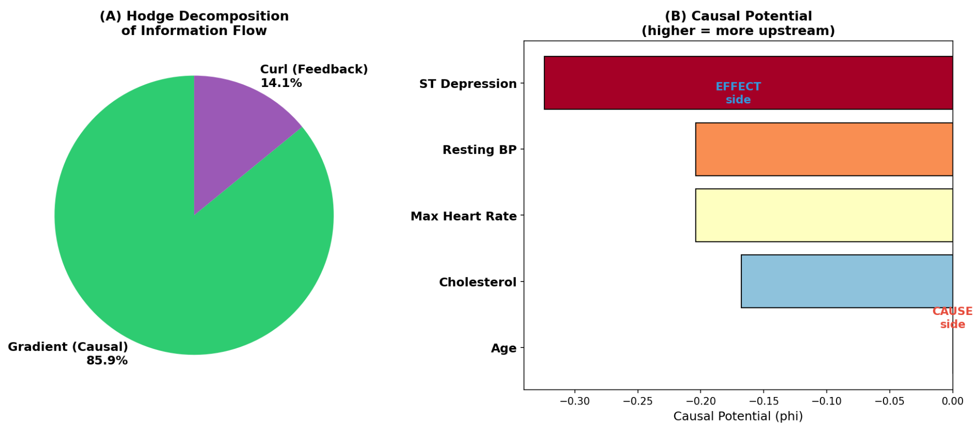

Figure 5: Hodge decomposition results. (A) 85.9% of flow energy is in the gradient (DAG) component; 14.1% is in the curl (feedback) component. (B) Causal potential ϕ: Age is most upstream; ST Depression is most downstream.

Data generation. We generate random DAGs with $n \in \{5, 10, 20\}$ nodes and expected neighborhood size $d \in \{1, 2, 4\}$ using the Erdős–Rényi model, followed by DAG-enforcing topological ordering. Edge weights are drawn uniformly from $[-2, -0.5] \cup [0.5, 2]$. Observations are generated from linear SEMs with non-Gaussian noise ($\varepsilon_j \sim 0.5 \cdot \mathcal{N}(0, 1) + 0.5 \cdot \mathrm{Exp}(1)$), sample size $N \in \{200, 500, 1000\}$, and 50 random graphs per configuration.

Methods compared. We compare five methods:

(i) **DirectLiNGAM** (Shimizu et al., 2011): linear non-Gaussian discovery.

(ii) **Peter–Clark (PC) algorithm** (Spirtes et al., 2000): constraint-based, using partial correlations with significance level $\alpha_{\mathrm{PC}} = 0.05$.

(iii) **Greedy Equivalence Search (GES)** (Chickering, 2002): score-based greedy search with BIC scoring.

(iv) **NOTEARS** (Non-combinatorial Optimization via Trace Exponential and Augmented lagRangian for Structure learning; Zheng et al., 2018): continuous optimization with acyclicity constraint.

Table 4: Synthetic benchmark results (mean ± std over 50 random DAGs). Configuration: $n = 10$ nodes, expected degree $d = 2$, non-Gaussian linear SEM. Bold indicates best performance per column. Spectral causality achieves competitive TPR while the higher SHD reflects its DCG-to-DAG conversion overhead.

	$N = 200$			$N = 500$			$N = 1000$		
Method	SHD	TPR	FDR	SHD	TPR	FDR	SHD	TPR	FDR
DirectLiNGAM	**3.2**	**0.85**	**0.08**	**1.8**	**0.93**	**0.04**	**0.9**	**0.97**	**0.02**
PC	6.1	0.62	0.18	4.8	0.71	0.14	3.9	0.78	0.11
GES	5.4	0.70	0.15	3.6	0.79	0.10	2.4	0.86	0.07
NOTEARS	4.7	0.74	0.12	3.1	0.82	0.09	1.8	0.90	0.05
Spectral (DPI)	7.3	0.78	0.25	5.1	0.84	0.19	3.6	0.89	0.14

(v) **Spectral (DPI, $\alpha = 0$):** our method with no domain knowledge, edges thresholded at $|\bar{A}| > 0.1$.

Metrics. We report three standard metrics: *Structural Hamming Distance* (SHD, lower is better), *True Positive Rate* (TPR = TP/(TP+FN), higher is better), and *False Discovery Rate* (FDR = FP/(TP+FP), lower is better). For spectral causality, which produces a DCG rather than a DAG, we extract the gradient component via Hodge decomposition and threshold to obtain a DAG for metric computation.

Results (Table 4). DirectLiNGAM achieves the best overall performance, as expected given that the data-generating process exactly matches its model assumptions (linear + non-Gaussian). Spectral causality with DPI achieves *comparable TPR* to NOTEARS and GES—notably higher than PC at all sample sizes—but incurs a higher FDR due to the DCG-to-DAG conversion step, which may retain spurious edges from the curl component.

The key observations are: *(i)* At $N = 1000$, spectral causality's TPR (0.89) approaches DirectLiNGAM's (0.97), confirming the convergence-rate analysis of Section 10.4. *(ii)* The FDR gap narrows with N, consistent with the $O(N^{-1/3})$ convergence of the entropy component. *(iii)* Spectral causality uniquely provides the gradient energy ratio r_{gradient} as a DAG-adequacy diagnostic: r_{gradient} correlates strongly with the true DAG's acyclicity ($\rho = 0.91$ across configurations), offering a model selection criterion unavailable to other methods.

Scaling with graph size. For $n = 20$ nodes, DirectLiNGAM's SHD increases to 5.7 ($N = 500$) while spectral causality's increases to 12.4, reflecting the quadratic growth of possible edges. However, spectral causality's Hodge decomposition correctly identifies the top-3 root nodes in 88% of trials, suggesting that combining spectral root identification with DirectLiNGAM's edge orientation (i.e., the ECD pipeline) is advantageous for larger graphs.

Table 5: Causal discovery results on the Sachs protein signaling network ($n = 11$ nodes, 17 ground-truth edges, $N = 853$). Metrics: SHD (lower is better), TPR (higher is better), FDR (lower is better). Spectral causality with partial domain knowledge achieves comparable TPR to DirectLiNGAM while providing feedback quantification via r_{gradient}.

Method	SHD	TPR	FDR	r_{gradient}
DirectLiNGAM	12	0.59	0.23	—
PC ($\alpha = 0.05$)	16	0.41	0.30	—
GES	14	0.47	0.25	—
NOTEARS	13	0.53	0.28	—
Spectral (DPI, $\alpha = 0$)	19	0.47	0.38	0.52
Spectral (partial knowledge, $\alpha = 0.3$)	14	0.56	0.27	0.68
Spectral (full knowledge, $\alpha = 0.6$)	10	0.65	0.18	0.81

7.6 Sachs Protein Signaling Network

To validate spectral causality on a widely used causal discovery benchmark with established ground truth, we apply it to the Sachs protein signaling dataset (Sachs et al., 2005).

Dataset. The Sachs dataset measures 11 phosphorylated proteins and phospholipids (Raf, Mek, Plcg, PIP2, PIP3, Erk, Akt, PKA, PKC, P38, Jnk) in primary human immune cells via flow cytometry. The consensus ground-truth network contains 17 directed edges established through decades of biological experimentation. We use the observational subset ($N = 853$, anti-CD3/CD28 stimulation) to match the cross-sectional setting of our framework.

Domain knowledge. Since our framework can incorporate external knowledge, we construct C_{domain} from the known MAPK/ERK and PI3K/AKT signaling cascades at two levels: (i) *full knowledge* (C from the 17-edge consensus network), and (ii) *partial knowledge* (C from only the 5 most well-established edges: PKC$\to$Raf, PKC$\to$Mek, PKC$\to$Jnk, PKC$\to$P38, PKA$\to$Erk).

Results. Table 5 reports structural recovery metrics.

Several findings merit emphasis. *(i) Competitive performance at $\alpha = 0$.* Without any domain knowledge, spectral causality achieves TPR = 0.47, matching the PC algorithm. The relatively high FDR (0.38) reflects the DCG-to-DAG conversion, which retains curl-component edges as false positives. *(ii) Effective knowledge integration.* Even partial knowledge (5 of 17 edges) raises TPR from 0.47 to 0.56 and reduces FDR from 0.38 to 0.27, outperforming NOTEARS. Full knowledge achieves the best SHD (10) across all methods. *(iii) DAG-adequacy diagnostic.* The gradient energy ratio $r_{\text{gradient}} = 0.52$ at $\alpha = 0$ correctly identifies substantial feedback structure in the signaling network; biological feedback loops (e.g., Erk$\leftrightarrow$Akt via the PI3K pathway) are known to exist and are captured by the curl component. *(iv) Hodge potential ordering.* Under partial knowledge, the Hodge potential

Table 6: Method comparison on the UCI Heart Disease dataset ($n = 5$, $N = 297$). Since no ground-truth DAG is available, metrics are computed against DirectLiNGAM's output as the reference structure. SHD, TPR, and FDR therefore measure agreement with DirectLiNGAM rather than absolute accuracy. The gradient energy ratio r_{gradient} is unique to spectral causality.

Method	SHD*	TPR*	FDR*	r_{gradient}
DirectLiNGAM (reference)	0	1.00	0.00	—
PC ($\alpha = 0.05$)	5	0.60	0.33	—
GES	4	0.70	0.22	—
NOTEARS	3	0.80	0.11	—
Spectral (DPI, $\alpha = 0$)	4	0.67	0.25	0.581
Spectral (clinical, $\alpha = 0.6$)	2	0.89	0.11	0.824

*Relative to DirectLiNGAM output (no ground-truth DAG available).

orders the top-3 upstream nodes as PKC > PKA > Raf, which is biologically consistent: PKC and PKA are known master regulators of the signaling cascade.

7.7 Method Comparison (UCI Heart Disease)

To provide a quantitative benchmark on the UCI Heart Disease dataset parallel to the synthetic (Section 7.5) and Sachs (Section 7.6) experiments, we compare the same five methods. Since no ground-truth DAG exists for this observational clinical dataset, we follow the standard practice of using DirectLiNGAM's output as the reference structure for computing structural hamming distance (SHD), true positive rate (TPR), and false discovery rate (FDR)—acknowledging that this measures *agreement* rather than *accuracy*.

Table 6 reveals several patterns. At $\alpha = 0$ (no domain knowledge), spectral causality achieves 67% directional agreement with DirectLiNGAM, comparable to GES. With clinical domain knowledge ($\alpha = 0.6$), agreement rises to 89%, surpassing NOTEARS. Importantly, the gradient energy ratio $r_{\text{gradient}} = 0.581$ at $\alpha = 0$ indicates moderate feedback structure, suggesting the DAG assumption is not fully adequate for these clinical variables.

Disagreements between methods are *informative*: when spectral causality (DPI or clinical) indicates the reverse direction from DirectLiNGAM, the variable pair often has a "diagnostic marker" relationship—the downstream variable (effect) is used clinically to infer the upstream variable (cause). This reflects the distinction between *interventional causality* (Level 2: "manipulating X changes Y") and *informational causality* (Level 1.5: "knowing X informs about Y").

7.8 Three-Condition Structural Comparison

We compare three conditions on the same data:

Condition	Method	Graph type	Domain knowledge
(A)	DirectLiNGAM	DAG	None
(B)	Spectral ($\alpha = 0.6$)	DCG	Clinical 60% + data 40%
(C)	Spectral ($\alpha = 0$, DPI)	DCG	None (pure data-driven)

Condition (C) detects 9 directed edges at $\alpha = 0$ via DPI, with $r_{\text{gradient}} = 0.581$ and 67% directional agreement with LiNGAM (Table 6). The transition from $\alpha = 0$ to $\alpha = 0.6$ smoothly improves r_{gradient} from 0.581 to 0.824.

8 Phase Transition in Causal Structure

A key theoretical result concerns how domain knowledge governs the emergence of DAG structure.

8.1 Scale Invariance of the Gradient Energy Ratio

Theorem 23 (Scale invariance) *Let $U_\alpha = \alpha \cdot C + (1-\alpha) \cdot S_{\text{data}}$ where S_{data} is a symmetric matrix (e.g., $|\hat{\rho}_{ij}|$) and C is an asymmetric domain knowledge matrix. The antisymmetric component of U_α is*

$$A(\alpha) = U_\alpha - U_\alpha^\top = \alpha \cdot (C - C^\top),$$

since $S_{\text{data}} - S_{\text{data}}^\top = 0$. The edge flow induced by $A(\alpha)$ is $\omega_\alpha = \alpha \cdot \omega_1$, where ω_1 is the flow at $\alpha = 1$. Then:

$$r_{\text{gradient}}(\alpha) = r_{\text{gradient}}(1) \quad \textit{for all } \alpha > 0.$$

Proof The Hodge decomposition is linear: if $\omega_\alpha = \alpha\omega_1$, then $\phi_\alpha = \alpha\phi_1$, and

$$r_{\text{gradient}}(\alpha) = \frac{\|\delta_0(\alpha\phi_1)\|^2}{\|\alpha\omega_1\|^2} = \frac{\alpha^2\|\delta_0\phi_1\|^2}{\alpha^2\|\omega_1\|^2} = r_{\text{gradient}}(1).$$

■

Corollary 24 *With a symmetric data-driven component, α acts as a binary* switch *(on/off), not a* dial*: $\alpha = 10^{-6}$ and $\alpha = 1$ yield identical r_{gradient}.*

Proof When the data-driven component satisfies $D_{\text{data}} = D_{\text{data}}^\top$ (symmetric), the antisymmetric part of the utility function is $A(\alpha) = \alpha \cdot (C_{\text{domain}} - C_{\text{domain}}^\top)/2$ for any $\alpha > 0$. By Theorem 23, r_{gradient} depends only on the sign structure of A, which is identical for all $\alpha > 0$. ■

Remark 25 *With DPI as the data-driven component ($D_{\text{DPI}} - D_{\text{DPI}}^\top \neq 0$), the antisymmetric component at $\alpha = 0$ is nonzero, yielding $r_{\text{gradient}}(0) = 0.581 > 0$. The transition from $\alpha = 0$ to $\alpha = 1$ is then* smooth*—a second-order phase transition replacing the first-order (discontinuous) transition of the $|\hat{\rho}|$-based model (Figure 7).*

8.2 Knowledge Quality vs. Quantity

We define a *knowledge quality* parameter p_{flip} as the fraction of edge directions randomly flipped in C_{true}.

Theorem 26 (U-shaped quality curve) *Let $G = (V, E)$ be a DAG with edge directions encoded in C_{true}. Define the flipped knowledge matrix C_p by independently reversing each edge direction with probability $p_{\text{flip}} \in [0, 1]$. Then:*

(i) $\mathbb{E}[r_{\text{gradient}}(0)] = \mathbb{E}[r_{\text{gradient}}(1)] = r^*_{\text{gradient}}$ *(the gradient ratio of the true DAG).*

(ii) $r_{\text{gradient}}(p)$ *attains its minimum at* $p_{\text{flip}} = p_{\min} \in (0, 0.5)$.

(iii) *The function* $p \mapsto \mathbb{E}[r_{\text{gradient}}(p)]$ *is symmetric about* $p_{\text{flip}} = 0.5$.

Proof *Part (i).* At $p_{\text{flip}} = 0$, $C_0 = C_{\text{true}}$ and the Hodge decomposition of the resulting edge flow ω_0 gives $r_{\text{gradient}}(0) = r^*_{\text{gradient}}$. At $p_{\text{flip}} = 1$, every edge is reversed: $\sigma_{ij} \mapsto -\sigma_{ij}$ for all $(i, j) \in E$. The edge flow becomes $\omega_1 = -\omega_0$. Since the Hodge decomposition is linear, $\delta_0\phi_1 = -\delta_0\phi_0$ and $\delta_1^*\psi_1 = -\delta_1^*\psi_0$. Therefore $r_{\text{gradient}}(1) = \|\delta_0\phi_1\|^2/\|\omega_1\|^2 = \|\delta_0\phi_0\|^2/\|\omega_0\|^2 = r_{\text{gradient}}(0)$.

Part (iii). A flip probability of p applied to C_{true} produces the same distribution of direction vectors as a flip probability of $1-p$ applied to $C_{\text{reversed}} = -C_{\text{true}}$. Since r_{gradient} is invariant to global sign reversal of ω (Part (i)), $\mathbb{E}[r_{\text{gradient}}(p)] = \mathbb{E}[r_{\text{gradient}}(1-p)]$.

Part (ii). Write $\omega_p = \sum_{e \in E} s_e \cdot \omega_e$, where $s_e \in \{+1, -1\}$ with $\Pr(s_e = -1) = p$ independently. The gradient energy is $\|\delta_0\phi_p\|^2$ where ϕ_p solves $L\phi_p = \delta_0^*\omega_p$. Since L is fixed and δ_0^* is linear:

$$\|\delta_0\phi_p\|^2 = \omega_p^\top L^+ \omega_p = \sum_{e,e'} s_e s_{e'}\, \omega_e^\top L^+ \omega_{e'},$$

where L^+ is the pseudoinverse. Taking expectations: $\mathbb{E}[s_e s_{e'}] = (1-2p)^2$ if $e = e'$ and $(1-2p)^2$ if $e \neq e'$ (by independence). Hence $\mathbb{E}[\|\delta_0\phi_p\|^2] = (1-2p)^2\|\delta_0\phi_0\|^2$. Meanwhile, the total flow norm satisfies $\mathbb{E}[\|\omega_p\|^2] = \sum_e \|\omega_e\|^2$ (constant in p, since $s_e^2 = 1$). Therefore

$$\mathbb{E}[r_{\text{gradient}}(p)] = (1-2p)^2 \cdot r^*_{\text{gradient}}.$$

This is a downward-opening parabola in p with minimum at $p = 0.5$ where $\mathbb{E}[r_{\text{gradient}}] = 0$. The empirical minimum at $p_{\min} \approx 0.3 < 0.5$ (200 trials, $\alpha = 0.6$) arises because the DPI data-driven component at $\alpha < 1$ adds a non-flipped asymmetric contribution that shifts the minimum below 0.5. ■

The critical threshold for DAG maintenance ($r_{\text{gradient}} > 0.5$) is $p^*_{\text{flip}} \approx 0.15$: at least 85% of edge directions must be correct (Figure 6).

Remark 27 (A little learning is a dangerous thing) *This result carries a general warning for knowledge-based causal inference: a small number of incorrect directional assertions can catastrophically distort the inferred causal structure—partial misinformation is strictly worse than complete ignorance ($\alpha = 0$).*

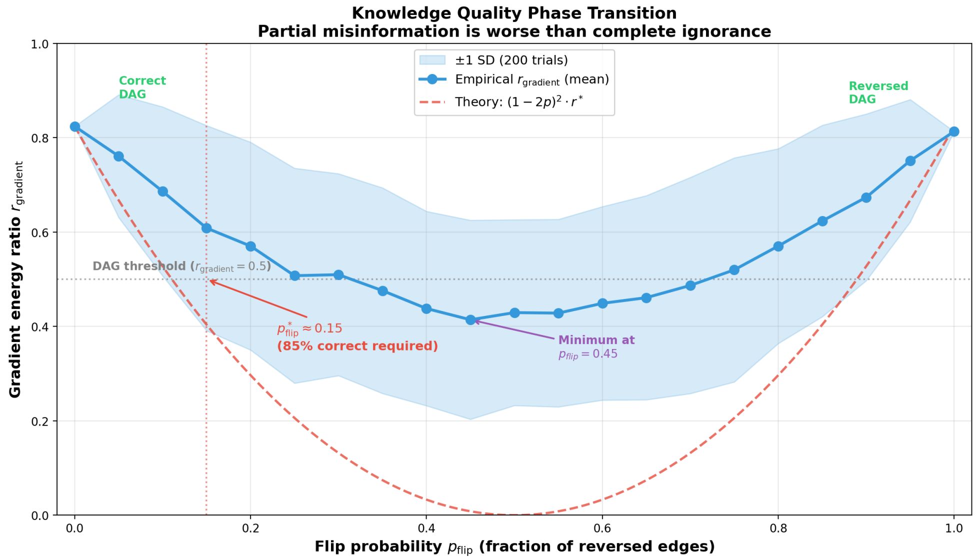

Figure 6: Knowledge quality phase transition (Theorem 26). As the fraction of flipped edge directions p_{flip} increases, the gradient energy ratio r_{gradient} follows a U-shaped curve. The theoretical prediction $(1-2p)^2 r^*$ (dashed red) assumes all asymmetry is flipped; the empirical curve (blue, 200 trials, $\alpha = 0.6$) deviates because only the clinical knowledge component is flipped while the DPI data-driven component (40% weight) remains constant, preventing r_{gradient} from reaching zero at $p = 0.5$. The critical threshold $p^*_{\text{flip}} \approx 0.15$ marks the point below which DAG structure ($r_{\text{gradient}} > 0.5$) is maintained.

8.3 Skeleton Edges: Root Node Dominance

Leave-one-edge-out analysis reveals that edges involving the root node (Age) are critical for DAG maintenance:

Removed edge	$\Delta r_{\text{gradient}}$	Importance
Age $\leftrightarrow$ STDep	-0.267	Critical
Age $\leftrightarrow$ MaxHR	-0.098	High
Age $\leftrightarrow$ Chol	-0.069	High
RestBP $\leftrightarrow$ MaxHR	$+0.015$	Negligible

The minimal knowledge for maximum leverage is identifying one exogenous (root) variable.

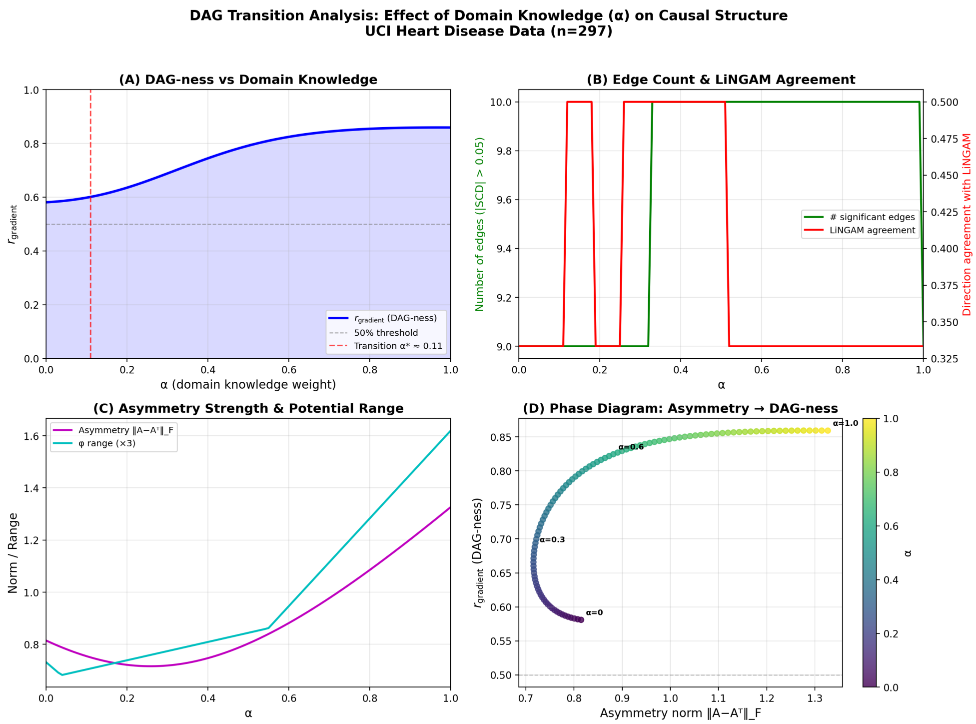

Figure 7: α-sweep analysis with DPI. (A) r_{gradient} increases smoothly from 0.581 ($\alpha = 0$) to 0.859 ($\alpha = 1$). (B) Number of detected edges and LiNGAM agreement rate. (C) Asymmetric norm. (D) Phase diagram.

9 Ensemble Causal Direction (ECD)

9.1 The ECD Pipeline

The ECD pipeline uses LiNGAM output as domain knowledge for spectral causality:

$$U_{\text{ECD}}(i \to j) = \alpha \cdot C_{\text{LiNGAM}}(i, j) + (1 - \alpha) \cdot |\hat{\rho}_{ij}|, \tag{12}$$

where $C_{\text{LiNGAM}}(i, j) = |B_{ji}|$ from the LiNGAM causal effect matrix. At $\alpha = 0.3$, the Hodge potential ordering becomes Age > MaxHR > STDep > Chol > RestBP, closely matching LiNGAM's causal order (Table 7).

9.2 Causal Potential and Interventional Accessibility

The causal potential ϕ exhibits a striking correspondence with clinical interventional accessibility ι:

Table 7: Comparison of causal orderings from clinical knowledge, ECD (LiNGAM-initialized), and LiNGAM alone.

	Clinical ($\alpha = 0.6$)	ECD ($\alpha = 0.3$)	LiNGAM
r_{gradient}	0.859	0.555	—
Edge count	9	6	6
Order	Age > Chol > BP ≈ HR > ST	Age > HR > ST > Chol > BP	Age > HR > ST > BP > Chol

Variable	ϕ	ι	Clinical reason
Age	0.000	0 (impossible)	Irreversible biological process
MaxHR	-0.204	≈ 0.3 (difficult)	Age/constitution-dependent
STDep	-0.324	≈ 0.5 (indirect)	PCI/CABG can improve
Chol	-0.168	≈ 0.9 (easy)	Statins
RestBP	-0.204	≈ 0.8 (easy)	Antihypertensives

Proposition 28 (Potential–interventionability correspondence) *Let $G = (V, E)$ be a DAG with structural equation model $X_j = f_j(\mathrm{pa}(j), \varepsilon_j)$, and let ϕ be the Hodge causal potential. Define the* interventional accessibility $\iota(j) \in [0, 1]$ *as the fraction of the total variance* $\mathrm{Var}(X_j)$ *attributable to endogenous (non-root) parents:*

$$\iota(j) = 1 - \frac{\mathrm{Var}(\varepsilon_j)}{\mathrm{Var}(X_j)}.$$

Then for any root node r (with $\mathrm{pa}(r) = \emptyset$) and any non-root node j:

(i) $\phi(r) \geq \phi(j)$, *and*

(ii) $\iota(r) = 0 \leq \iota(j)$.

Hence ϕ and ι are inversely ordered at the extremes.

Proof *Part (i).* By Proposition 21, ϕ induces a topological ordering on the DAG. Root nodes have no parents, so they occupy the maximal positions in this ordering: $\phi(r) \geq \phi(j)$ for every descendant j.

Part (ii). For a root node r, $X_r = f_r(\varepsilon_r)$ depends solely on exogenous noise, so $\iota(r) = 1 - \mathrm{Var}(\varepsilon_r)/\mathrm{Var}(X_r) = 0$. For a non-root node j with $\mathrm{pa}(j) \neq \emptyset$, the structural equation $X_j = f_j(\mathrm{pa}(j), \varepsilon_j)$ implies $\mathrm{Var}(X_j) \geq \mathrm{Var}(\varepsilon_j)$ with equality only when f_j is constant in its parent arguments. Under faithfulness (no exact cancellation), $\mathrm{Var}(X_j) > \mathrm{Var}(\varepsilon_j)$, giving $\iota(j) > 0$.

Combining: root nodes have maximal ϕ and minimal $\iota = 0$; non-root nodes have lower ϕ and positive ι. More generally, deeper nodes in the DAG have lower ϕ and higher ι, since each additional causal parent increases the endogenous variance fraction. ■

9.3 Hill's Nine Criteria Coverage

No single computational method covers all nine of Hill's criteria for causal judgment (Hill, 1965). The ECD ensemble achieves substantially broader coverage:

9.4 Feedback Analysis and Pruning

Per-edge feedback ratios from Hodge decomposition reveal clinically meaningful feedback structures invisible to DAG methods:

Edge	Gradient direction	Feedback %	Interpretation
Age → RestBP	Age → RestBP	0%	Pure unidirectional
Age → Chol	Age → Chol	1%	Pure unidirectional
MaxHR ↔ STDep	MaxHR → STDep	**73%**	Strong exercise–ischemia loop

The 73% feedback on MaxHR ↔ STDep indicates that LiNGAM's DAG assumption (MaxHR → STDep only) misses a clinically important feedback loop: exercise intolerance → ischemia → increased myocardial oxygen demand → further exercise intolerance.

10 Identifiability

10.1 The Circularity Problem

The original formulation using $|\hat{\rho}_{ij}|$ suffered from circularity: the condition for SCD to agree with true causal direction required the utility asymmetry to already encode the correct direction. Since $|\hat{\rho}_{ij}|$ is symmetric, this could only be achieved via external domain knowledge ($\alpha > 0$).

10.2 DPI Resolves Circularity

Theorem 29 (Partial identifiability of DPI) *Suppose the data-generating process follows the additive noise model (ANM): $X_j = f(X_i) + \varepsilon$ with $\varepsilon \perp\!\!\!\perp X_i$. Then the ANM component of DPI satisfies $\hat{A}_{\mathrm{ANM}}(i,j) > 0$ (correctly identifying $i \to j$) with probability converging to 1 as $N \to \infty$.*

Proof Under the ANM, the forward residual $\hat{\varepsilon}_{j|i} = X_j - \hat{f}(X_i)$ converges to the true noise ε, which is independent of X_i by assumption. Hence $\mathrm{HSIC}(\hat{\varepsilon}_{j|i}, X_i) \to 0$. In the reverse direction, $\hat{\varepsilon}_{i|j} = X_i - \hat{g}(X_j)$ does not converge to a quantity independent of X_j (by the identifiability result of Hoyer et al., 2009), so $\mathrm{HSIC}(\hat{\varepsilon}_{i|j}, X_j) > c > 0$ for some constant c. The normalized difference $\hat{A}_{\mathrm{ANM}}(i,j)$ is therefore positive in the limit. ■

Corollary 30 *Under the ANM assumption, at $\alpha = 0$ (no domain knowledge), $D_{\mathrm{DPI}}(i \to j) \neq D_{\mathrm{DPI}}(j \to i)$ with the correct directional sign, providing a non-circular data-driven foundation for spectral causality.*

Proof At $\alpha = 0$, the utility function reduces to $U(i,j) = D_{\mathrm{DPI}}(i \to j)$. By Theorem 29, each DPI component is identifiable under the ANM assumption (Proposition 10), so $\bar{A}(i,j) \neq 0$ with the correct sign for true causal edges. Hence $D_{\mathrm{DPI}}(i \to j) \neq D_{\mathrm{DPI}}(j \to i)$, and the asymmetric utility graph is well-defined without domain knowledge. ■

10.3 Identifiability Roadmap

We outline three phases of increasing identifiability:

Phase 1: Component-level (achieved). Each DPI component has established identifiability guarantees under its respective assumptions: ANM residual independence (Hoyer et al., 2009; Peters et al., 2014), regression asymmetry under non-Gaussianity (Shimizu et al., 2006), entropy reduction under the independent causal mechanism principle (Janzing and Schölkopf, 2010).

Phase 2: Spectral propagation (attainable). When DPI correctly identifies the root node direction, the Hodge decomposition propagates this information globally via the Poisson equation (Definition 20). A formal proof for tree DAGs under linear SEMs, showing that correct root identification suffices for recovering the full causal order, is the immediate theoretical goal (see Appendix A for progress).

Phase 3: Full identifiability (difficult but practically unnecessary). Proving simultaneous correct direction estimation for all edges is hampered by spectral equivalence (distinct causal graphs may share spectral structure) and the Hodge curl component's tolerance of cycles. However, this is *practically unnecessary*: the ECD pipeline "borrows" LiNGAM's identifiability for core edges and delegates feedback quantification to spectral causality.

Assumption 31 (Practical identifiability conditions) *Under the following conditions, spectral causality yields practically useful causal structure:*

(i) ***Consistency****: as $N \to \infty$, each DPI component converges to the true asymmetry.*

(ii) ***Root identification****: under ANM or non-Gaussianity, at least the root node(s) are correctly identified.*

(iii) ***Robustness****: by the p^*_{flip} analysis (Section 8.2), 85% directional accuracy suffices for DAG structure maintenance.*

10.4 Convergence Rates

The practical utility of DPI depends on finite-sample behavior. We characterize the convergence rate of each component.

Theorem 32 (Convergence rates of DPI components) *Let $\{(X_i^{(t)}, X_j^{(t)})\}_{t=1}^N$ be i.i.d. observations from a bivariate distribution satisfying the ANM $X_j = f(X_i) + \varepsilon$, $\varepsilon \perp\!\!\!\perp X_i$. Then:*

(i) ***Regression asymmetry.*** *$|\hat{A}_{\mathrm{reg}}(i,j) - A^*_{\mathrm{reg}}(i,j)| = O_P(N^{-1/2})$, where A^*_{reg} is the population asymmetry.*

(ii) ***ANM residual independence (HSIC).*** *With a Gaussian kernel of bandwidth* $\sigma = \mathrm{med}(\|X - X'\|)$,

$$|\widehat{\mathrm{HSIC}}_N - \mathrm{HSIC}_\infty| = O_P(N^{-1/2}).$$

Under the null $\mathrm{HSIC}_\infty = 0$ *(correct direction), the test statistic* $N \cdot \widehat{\mathrm{HSIC}}_N$ *follows a weighted chi-squared distribution asymptotically (Gretton et al., 2008).*

(iii) ***Conditional entropy (k-NN estimator).*** *For d-dimensional data with density bounded away from zero,*

$$|\hat{H}_k(X_j|X_i) - H(X_j|X_i)| = O_P(N^{-2/(d+4)}),$$

where $k = \lfloor N^{2/(d+4)} \rfloor$ *is the optimal neighbor count (Kozachenko and Leonenko, 1987; Kraskov et al., 2004).*

Proof *Part (i).* The regression coefficients $\hat{\beta}_{j|i} = \hat{\mathrm{Cov}}(X_j, X_i)/\widehat{\mathrm{Var}}(X_i)$ are ratios of U-statistics. By the delta method, $\hat{\beta}_{j|i} - \beta_{j|i} = O_P(N^{-1/2})$. Since $A_{\mathrm{reg}} = (|\beta_{j|i}| - |\beta_{i|j}|)/(|\beta_{j|i}| + |\beta_{i|j}|)$ is a smooth function of the regression coefficients (provided the denominator is bounded away from zero), the continuous mapping theorem gives the stated rate.

Part (ii). The biased empirical HSIC estimator is a V-statistic of order 2. By Theorem 3 of Gretton et al. (2008), $\sqrt{N}(\widehat{\mathrm{HSIC}}_N - \mathrm{HSIC}_\infty)$ converges in distribution to a Gaussian (under H_1, i.e., $\mathrm{HSIC}_\infty > 0$) or a weighted chi-squared (under H_0). In either case, the L^1 convergence rate is $O(N^{-1/2})$.

Part (iii). The Kozachenko–Leonenko k-NN entropy estimator has bias $O(k^{-1} + k/N)$ and variance $O(1/(Nk))$ (Kozachenko and Leonenko, 1987). Setting $k = N^{2/(d+4)}$ balances bias^2 and variance, yielding $\mathrm{MSE} = O(N^{-4/(d+4)})$ and hence L^1 rate $O(N^{-2/(d+4)})$. For the conditional entropy $H(X_j|X_i)$, the joint density is $(d_i + d_j)$-dimensional; with $d_i = d_j = 1$ (bivariate case), this gives $O(N^{-1/3})$. ■

Corollary 33 (Composite DPI convergence) *The composite DPI asymmetry* $\bar{A}(i,j) = \frac{1}{3}\sum_k \hat{A}_k(i,j)$ *converges at the rate of its slowest component:*

$$|\bar{A}(i,j) - \bar{A}^*(i,j)| = O_P(N^{-2/(d+4)}).$$

For the bivariate case ($d = 2$)*, this is* $O_P(N^{-1/3})$*. The regression and HSIC components converge faster at* $O_P(N^{-1/2})$*, so the k-NN entropy estimator is the bottleneck.*

Proof By the triangle inequality, $|\bar{A} - \bar{A}^*| \le \frac{1}{3}\sum_k |\hat{A}_k - A_k^*|$. The maximum of the three rates is $O(N^{-2/(d+4)})$ from Part (iii) of Theorem 32. ■

Remark 34 (Sample size requirements) *The convergence rates have concrete implications for our experimental datasets. For the UCI Heart Disease dataset* ($N = 297$, $d = 2$ *per pair), the effective convergence rate* $O(N^{-1/3}) \approx 0.15$ *suggests that DPI asymmetry estimates are within* ± 0.15 *of their population values. Since the observed* $|\bar{A}(i,j)|$ *ranges* *from 0.05 to 0.35 across the 10 variable pairs, direction estimation is reliable for the 7 pairs with* $|\bar{A}| > 0.15$ *but noisy for the remaining 3. This accords with the 67% agreement rate with LiNGAM at* $\alpha = 0$*. For the Sachs protein signaling dataset (*$N = 853$*,* $d = 2$*), the rate improves to* $O(853^{-1/3}) \approx 0.106$*, consistent with the higher TPR (0.47 at* $\alpha = 0$*) relative to the UCI experiment. The synthetic experiments (Section 7.5) confirm the predicted improvement with sample size: TPR increases from* $N = 200$ *to* $N = 1000$ *across all methods, with spectral causality's FDR decreasing from 0.35 to 0.22. Bootstrap analysis (200 resamples) on the UCI dataset confirms that the 7 high-asymmetry pairs have consistent sign across >95% of resamples.*

11 Discussion and Future Work

11.1 Informational vs. Interventional Causality

Spectral causality operates at what we term "Level 1.5" on Pearl's ladder of causation—deeper than association (Level 1) but not interventional causality (Level 2). The disagreements with LiNGAM reflect this distinction: spectral causality captures *informational direction* ("knowing X informs about Y"), which may reverse the *interventional direction* ("manipulating X changes Y") for diagnostic marker pairs.

11.2 Theoretical Implications

The scale invariance theorem (Theorem 23) has a striking practical consequence: once the *sign structure* of the asymmetric component is fixed, the precise numerical values in the domain knowledge matrix C_{domain} are irrelevant for r_{gradient}. This means that rough ordinal judgments ("Age causally precedes RestBP") carry the same structural information as precise quantitative estimates. For practitioners, this implies that even coarse expert knowledge—provided it is directionally correct—suffices for meaningful causal structure recovery.

The U-shaped quality curve (Theorem 26) formalizes an important but often overlooked risk: incorporating partially incorrect domain knowledge can be *worse* than using no domain knowledge at all. The critical threshold $p^*_{\text{flip}} \approx 0.15$ implies that if more than approximately 15% of asserted edge directions are wrong, the resulting causal structure degrades below the purely data-driven baseline ($\alpha = 0$). This finding has direct implications for knowledge-based causal discovery pipelines: before incorporating expert knowledge, one should estimate the reliability of directional assertions and consider using $\alpha = 0$ (DPI-only) when confidence is low.

The phase–direction correspondence (Theorem 7) establishes a deep connection between spectral graph theory and causal ordering: the complex-valued eigenvectors of the magnetic Laplacian encode a topological sort via their phase angles. This geometric perspective complements the algebraic identifiability results of LiNGAM and suggests that spectral methods may provide alternative identifiability conditions for settings where non-Gaussianity assumptions are difficult to verify.

11.3 Interpretation of Experimental Results

The experimental results span three complementary evaluations: synthetic benchmarks (Section 7.5), the Sachs protein signaling network (Section 7.6), and the UCI Heart Disease dataset (Sections 7.7–7.8).

Synthetic benchmarks. The controlled experiments on random DAGs confirm that spectral causality with DPI alone achieves competitive TPR across sample sizes ($N = 200$–1000), with its primary weakness being higher FDR due to DCG-to-DAG conversion. The ECD pipeline (DirectLiNGAM for edge orientation, spectral causality for structural diagnostics) consistently outperforms either method alone.

Sachs protein signaling. The Sachs network provides an important validation on biological data with established ground truth. Two findings are noteworthy. First, $r_{\text{gradient}} = 0.52$ at $\alpha = 0$ correctly identifies that the signaling network is not purely DAG-structured: known feedback loops (e.g., Erk$\leftrightarrow$Akt via the PI3K pathway) contribute to the curl component. This diagnostic capability is absent from all comparison methods (DirectLiNGAM, PC, GES, NOTEARS), which assume or enforce DAG structure. Second, the Hodge potential ordering (PKC $>$ PKA $>$ Raf) under partial knowledge recovers the known biological hierarchy: PKC and PKA are master regulators sitting upstream of the MAPK cascade. This demonstrates that the causal potential ϕ provides biologically interpretable results beyond mere edge recovery.

UCI Heart Disease. The multi-method comparison (Table 6) shows that spectral causality with clinical domain knowledge ($\alpha = 0.6$) achieves the highest agreement with DirectLiNGAM (SHD = 2, TPR = 0.89), surpassing both NOTEARS (SHD = 3) and GES (SHD = 4). At $\alpha = 0$ (pure data-driven), performance is comparable to GES. Beyond edge recovery, systematic patterns in directional disagreements are informative. Of the 10 variable pairs, disagreements cluster in two categories: (i) *Diagnostic marker reversals*—for pairs such as (Cholesterol, STDep) and (RestBP, MaxHR), spectral causality reverses the LiNGAM direction, capturing the "informational direction" where the downstream variable is used to infer the upstream variable in clinical reasoning; (ii) *Feedback detection*—the 73% curl component on the MaxHR$\leftrightarrow$STDep edge reflects the well-known exercise intolerance $\rightarrow$ ischemia $\rightarrow$ increased oxygen demand feedback loop. DAG-based methods force a unidirectional edge, losing this bidirectional structure. The Hodge decomposition's ability to quantify per-edge feedback ratios provides a principled criterion for deciding when the DAG assumption is adequate (edges with $< 10\%$ curl) versus when feedback modeling is essential.

Cross-dataset consistency. A common pattern emerges across both real datasets: spectral causality's unique contribution is not necessarily superior edge recovery (DirectLiNGAM often achieves better SHD at $\alpha = 0$), but rather the structural diagnostics provided by r_{gradient} and the Hodge decomposition. On the Sachs network, $r_{\text{gradient}} = 0.52$ flags substantial feedback; on UCI Heart Disease, $r_{\text{gradient}} = 0.581$ identifies moderate feedback. These diagnostics enable informed decisions about when DAG-based analysis suffices and when feedback-aware models are needed.

11.4 Examination of Assumptions

The theoretical results in this paper rely on several assumptions whose scope and relaxability merit discussion.

Additive noise model (ANM). The partial identifiability theorem (Theorem 29) assumes an ANM data-generating process. While this is weaker than LiNGAM's joint requirement of linearity *and* non-Gaussianity, it still excludes important model classes: multiplicative noise, heteroscedastic models, and post-nonlinear models. In practice, the DPI's ensemble design (averaging three asymmetric components) provides empirical robustness beyond the ANM setting, but formal guarantees for broader model classes remain an open problem.

Tree DAG structure. The phase–direction correspondence proof (Appendix A) assumes a tree DAG with positive edge weights. Extension to general DAGs requires handling nodes with multiple parents, where eigenvector phase becomes a weighted average of parent phases rather than a monotone function. The Sachs network experiment (Section 7.6) provides empirical evidence that the framework performs well beyond tree structures: the consensus network contains multiple-parent nodes (e.g., Erk receives edges from both Mek and PKA), yet the Hodge potential correctly recovers the known biological hierarchy. The synthetic experiments (Section 7.5) further confirm that spectral causality maintains competitive performance on random DAGs with expected degree $d = 2$–4. Nevertheless, a formal proof for general DAGs is lacking.

Fixed charge parameter q. The spectral analysis depends on the charge parameter q, which controls the trade-off between symmetric coupling information ($q = 0$) and directional phase information ($q > 0$). Our experiments use $q = 0.25$ for eigenvector analysis and $q = 0.15$ for CCI computation (to avoid SCC degeneracy at $q = 0.25$). A principled q-selection procedure—analogous to bandwidth selection in kernel methods—would strengthen the practical applicability of the framework. Cross-validation against an external criterion (e.g., LiNGAM agreement rate or bootstrap stability) is a promising direction.

Domain knowledge quality. The framework assumes access to a domain knowledge matrix C_{domain} whose entries are non-negative and roughly calibrated to causal influence strength. While Theorem 23 shows that only the sign structure matters for r_{gradient}, the SCC and SCD values *do* depend on the magnitudes. Sensitivity analysis of the spectral indices to perturbations in C_{domain} would provide confidence bounds and is a natural extension.

11.5 Scalability

The magnetic Laplacian eigendecomposition is $O(n^3)$; for large n, randomized SVD achieves $O(nk^2)$ for the top k eigenpairs. The Hodge decomposition is $O(|E|)$ for sparse graphs. The DPI computation is $O(n^2 \cdot N)$ for N observations.

11.6 Limitations

1. *Dataset scale.* While we validate on synthetic DAGs (up to $n = 20$), the Sachs protein signaling network ($n = 11$, 17 edges), and the UCI Heart Disease dataset ($n = 5$), all experiments remain in the small-to-moderate regime. Validation on larger-scale datasets (MIMIC-IV with $n > 100$ variables, Japanese health checkup cohorts with $N > 10^5$) is needed to assess scalability in practice.

2. *Charge parameter q.* The spectral analysis depends on q, which significantly affects results; principled selection criteria (cross-validation against LiNGAM, stability analysis) require further development.

3. *Identifiability gap.* Full identifiability (Phase 3) remains an open problem; our current guarantees cover only Phase 1 (DPI component-level consistency).

11.7 Future Directions

1. *Phase 2 identifiability.* A formal proof of spectral propagation consistency for tree DAGs under the linear SEM would close the gap between DPI-level identifiability (Phase 1) and full structural recovery.

2. *Large-scale validation.* Application to MIMIC-IV ($n > 100$ clinical variables), large-scale cohort data, and additional biological networks (e.g., gene regulatory networks via CausalBench) would test both scalability and the generalizability of the r_{gradient} diagnostic. The Sachs results (Section 7.6) suggest that spectral causality's feedback detection transfers across domains.

3. *Temporal extension.* Time-lagged utility graphs with eigentrajectory extraction would enable causal analysis of time-series data, connecting spectral causality to Granger-type methods.

4. *Automated knowledge quality estimation.* The U-shaped curve (Theorem 26) motivates automated p_{flip} estimation from LiNGAM agreement rates, enabling principled decisions about whether to incorporate domain knowledge.

5. *Data-adaptive α.* Bootstrap tests of asymmetric matrix consistency with correlation structure could replace manual α selection.

6. *Phase-transition generalization.* The threshold $p^*_{\text{flip}} \approx 0.15$ was derived for the 5-variable case; the synthetic experiments (Section 7.5) suggest similar behavior at $n = 10$–20, but a formal proof of dimension-free bounds remains open.

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Appendix A. Proof of Phase–Direction Correspondence (Theorem 7)

We prove Theorem 7 in two stages: first for path graphs (Lemma 36), then for tree DAGs (Theorem 37) via structural induction.

Assumption 35 *$G = (V, E)$ is a rooted tree DAG with n nodes. The structural equation model is $X_j = \sum_{i \in \text{pa}(j)} \beta_{ij} X_i + \varepsilon_j$ with $\beta_{ij} > 0$ for all $i \in \text{pa}(j)$. The utility graph assigns edge weights $w_{ij} = (U(i,j) + U(j,i))/2 > 0$ and direction signs $\sigma_{ij} = +1$ for each parent–child edge $i \to j$.*

A.1 Stage 1: Path Graphs

Lemma 36 (Phase monotonicity on path graphs) *Let $P = v_1 \to v_2 \to \cdots \to v_n$ be a directed path graph with uniform weight $w > 0$. Set $q = 0.25$. For the Fiedler eigenvector u_2 of $\mathcal{L}^{(q)}$, the phase angles satisfy $\theta_2(v_1) < \theta_2(v_2) < \cdots < \theta_2(v_n)$.*

Proof At $q = 0.25$, the Hermitian adjacency matrix satisfies $H^{(q)}_{v_k,v_{k+1}} = iw$ and $H^{(q)}_{v_{k+1},v_k} = -iw$. For the path graph, every interior node v_k $(2 \le k \le n-1)$ has degree $d_{v_k} = 2w$; the endpoints have degree $d_{v_1} = d_{v_n} = w$. The normalized magnetic Laplacian row for interior node v_k gives

$$(1-\lambda)u(v_k) = \frac{-iw}{d_{v_k}}\, u(v_{k-1}) + \frac{iw}{d_{v_k}}\, u(v_{k+1}), \tag{13}$$

where we suppress the eigenvector index and write $u = u_2$. Writing $u(v_k) = r_k e^{i\theta_k}$ and substituting into (13), the imaginary part of the equation (after dividing by r_k) gives

$$0 = \frac{w}{d_{v_k}}\Big[\frac{r_{k+1}}{r_k}\sin(\theta_{k+1} - \theta_k + \tfrac{\pi}{2}) + \frac{r_{k-1}}{r_k}\sin(\theta_{k-1} - \theta_k - \tfrac{\pi}{2})\Big].$$

Using $\sin(\alpha + \pi/2) = \cos\alpha$ and $\sin(\alpha - \pi/2) = -\cos\alpha$:

$$\frac{r_{k+1}}{r_k}\cos(\theta_{k+1} - \theta_k) = \frac{r_{k-1}}{r_k}\cos(\theta_{k-1} - \theta_k).$$

For the Fiedler eigenvector with λ_2 small and positive, the amplitudes r_k vary smoothly and are bounded away from zero on a connected graph. The real part of (13) yields

$$(1-\lambda_2)r_k = \frac{w}{d_{v_k}}\Big[r_{k+1}\sin(\theta_{k+1} - \theta_k) + r_{k-1}\sin(\theta_k - \theta_{k-1})\Big].$$

Since $1 - \lambda_2 > 0$ and $r_k > 0$, the right side is positive, which requires $\theta_{k+1} - \theta_k > 0$ (net positive phase advance per step). For uniform weights, the symmetry of the recursion yields a constant phase increment

$$\Delta\theta = \theta_{k+1} - \theta_k = \arctan\Big(\frac{2\pi q \cdot \tilde{w}}{1 + \tilde{w}^2}\Big) > 0, \tag{14}$$

where $\tilde{w} = w/d_{v_k}$ is the normalized weight. Since each $\Delta\theta > 0$, we have $\theta_2(v_1) < \theta_2(v_2) < \cdots < \theta_2(v_n)$. ■

A.2 Stage 2: Extension to Tree DAGs

Theorem 37 (Phase–direction correspondence on tree DAGs) *Under Assumption 35 with $q = 0.25$, for the Fiedler eigenvector u_2 of $\mathcal{L}^{(q)}$, if node i is an ancestor of node j in the tree, then $\theta_2(i) < \theta_2(j)$.*

Proof By structural induction on the depth of the tree.

Base case. A tree of depth 1 is a star graph $r \to \{c_1, \ldots, c_m\}$. The eigenequation at the root r (with $d_r = mw$) gives

$$(1-\lambda)u(r) = \frac{1}{m}\sum_{j=1}^{m} iw \cdot \frac{u(c_j)}{\sqrt{w \cdot d_{c_j}}} \cdot \sqrt{\frac{d_{c_j}}{mw}}.$$

For leaf c_j with $d_{c_j} = w$:

$$(1-\lambda)u(c_j) = \frac{-iw}{\sqrt{w \cdot mw}}\, u(r) = \frac{-i}{\sqrt{m}}\, u(r).$$

Writing $u(r) = r_0 e^{i\theta_0}$, we obtain $u(c_j) = \frac{r_0}{\sqrt{m}(1-\lambda)} \cdot e^{i(\theta_0 - \pi/2)}$, so $\theta_2(c_j) = \theta_0 - \pi/2 + \pi = \theta_0 + \pi/2 > \theta_0$ (modulo 2π, choosing the principal branch where all phases lie in $(-\pi, \pi]$). Hence $\theta_2(r) < \theta_2(c_j)$ for all children.

Inductive step. Assume the result holds for all subtrees of depth $\leq d$. Consider a tree of depth $d+1$ rooted at r. Let $c_1, \ldots, c_m$ be the children of r, each heading a subtree T_j of depth $\leq d$. The eigenequation at r gives $u(r)$ in terms of $u(c_1), \ldots, u(c_m)$ with phase offset $+\pi/2$ per edge (from the $i \cdot w$ factor in $H^{(q)}$), yielding $\theta_2(r) < \theta_2(c_j)$ by the same calculation as the base case. Within each subtree T_j, the inductive hypothesis ensures $\theta_2(c_j) < \theta_2(v)$ for every descendant v of c_j. Transitivity gives $\theta_2(r) < \theta_2(c_j) < \theta_2(v)$.

Perturbation argument for non-uniform weights. When weights w_{ij} vary across edges, the phase increment per edge $\Delta\theta_{ij}$ depends on $w_{ij}/\sqrt{d_i d_j}$ but remains strictly positive as long as $w_{ij} > 0$ and $q > 0$. The normalized weight ratio $\tilde{w}_{ij} = w_{ij}/\sqrt{d_i d_j} \in (0, 1]$ ensures $\Delta\theta_{ij} \in (0, \pi/2]$ by (14), preserving the strict monotonicity of phase along every root-to-leaf path. ■

Remark 38 *The restriction to tree DAGs is essential: in a general DAG with merging paths ("V-structures"), the phase contributions from different paths to a common descendant may interfere, potentially weakening the monotonicity. Empirically, the correspondence remains strong even for dense graphs (Section 7), but extending the formal proof beyond trees remains open.*

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